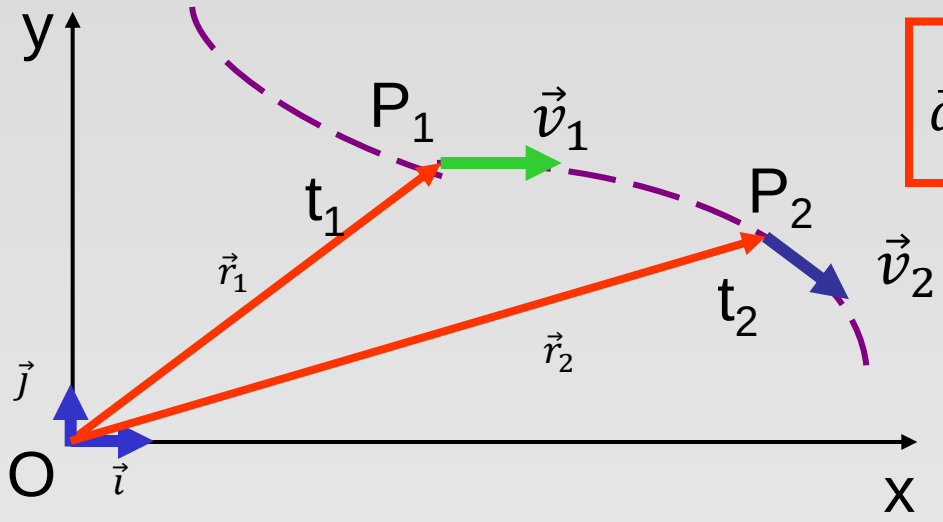
Física I

Turno I

Apuntes de Clase 4

Prof. Pedro Mendoza Zélis

Repaso:



Aceleración media

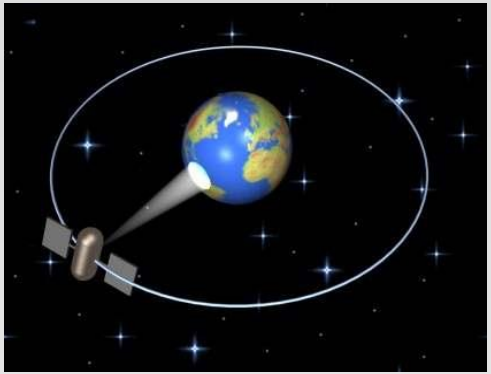
$$\boxed{\vec{a}_m = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}} \quad [m/s^2]$$

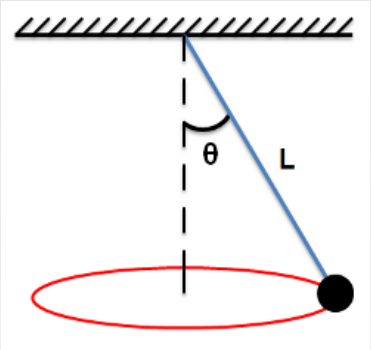
$$\Delta t = t_2 - t_1$$

Aceleración instantánea

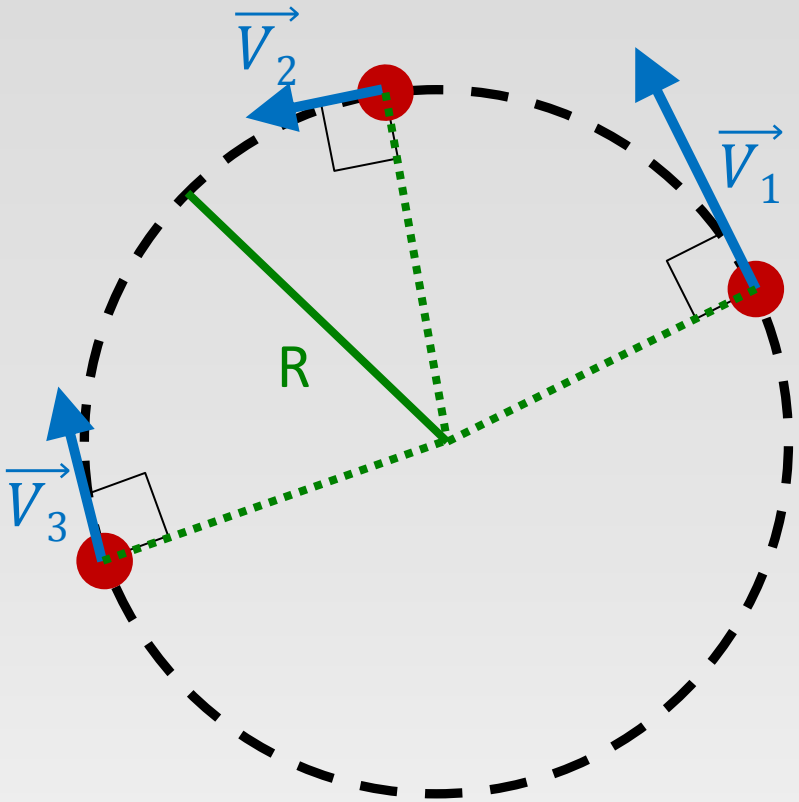
$$\boxed{\vec{a} = \lim_{\Delta t \to 0} \vec{a}_m = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}}$$

Movimiento Circular





Movimiento circular



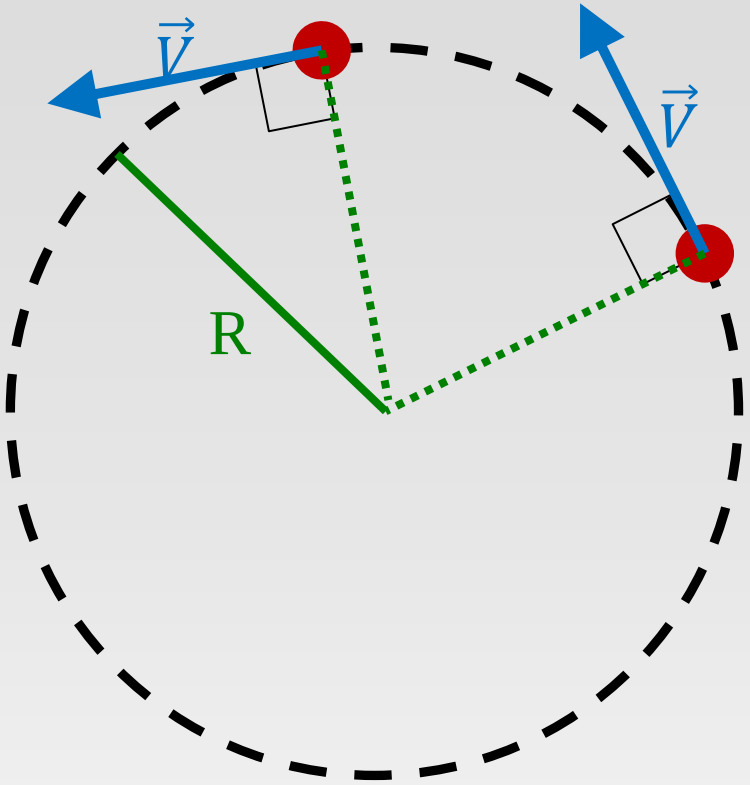
Movimiento circular uniforme

$|\vec{V}|$ = cte

T: período [seg]
tiempo empleado en dar una vuelta completa

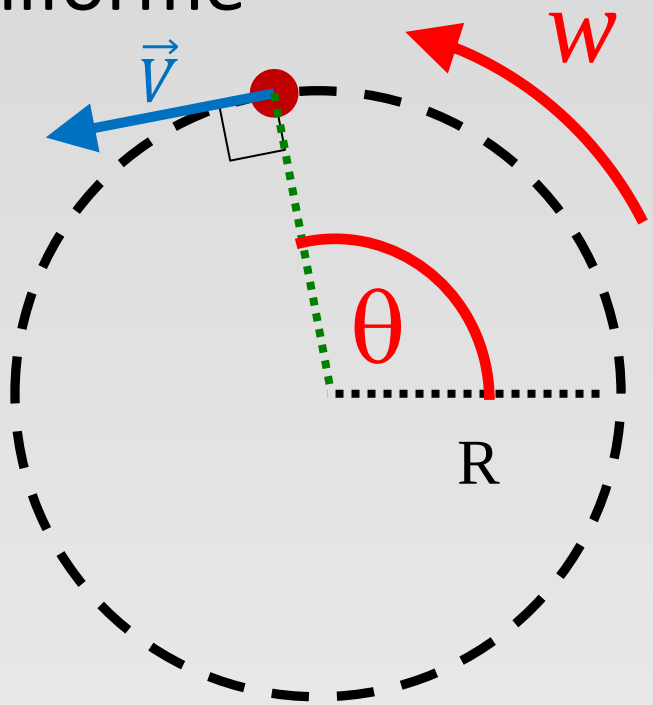
f: frecuencia [1/seg = Hz]
#vueltas/segundo

$$\frac{1}{f} = T$$



Movimiento circular uniforme

$|\vec{V}|$ = cte



w: velocidad angular [rad/seg]
radianes recorridos por segundo

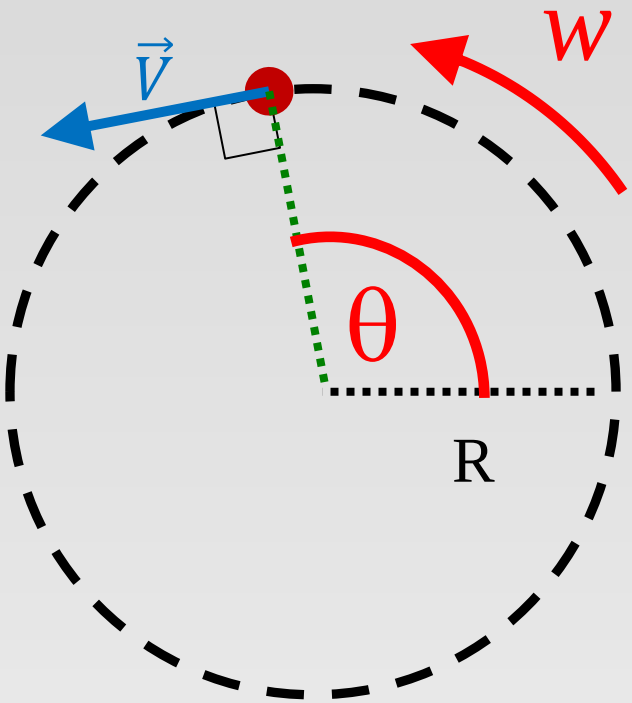
$$w = \frac{d\theta}{dt}$$

Relación entre T y *w*: $$w = \frac{2\,\pi}{T}$$

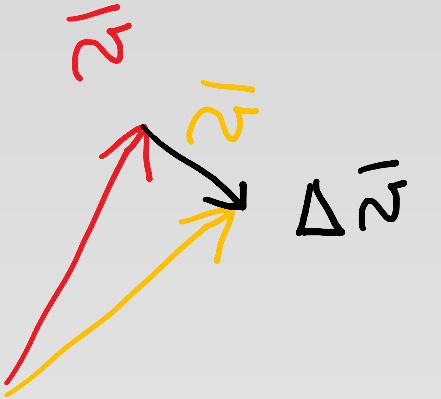
Movimiento circular

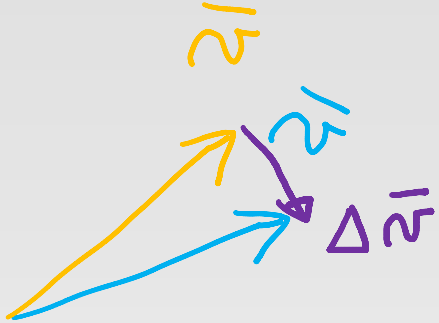
¿Cómo se relacionan $|\vec{V}|$ y w?

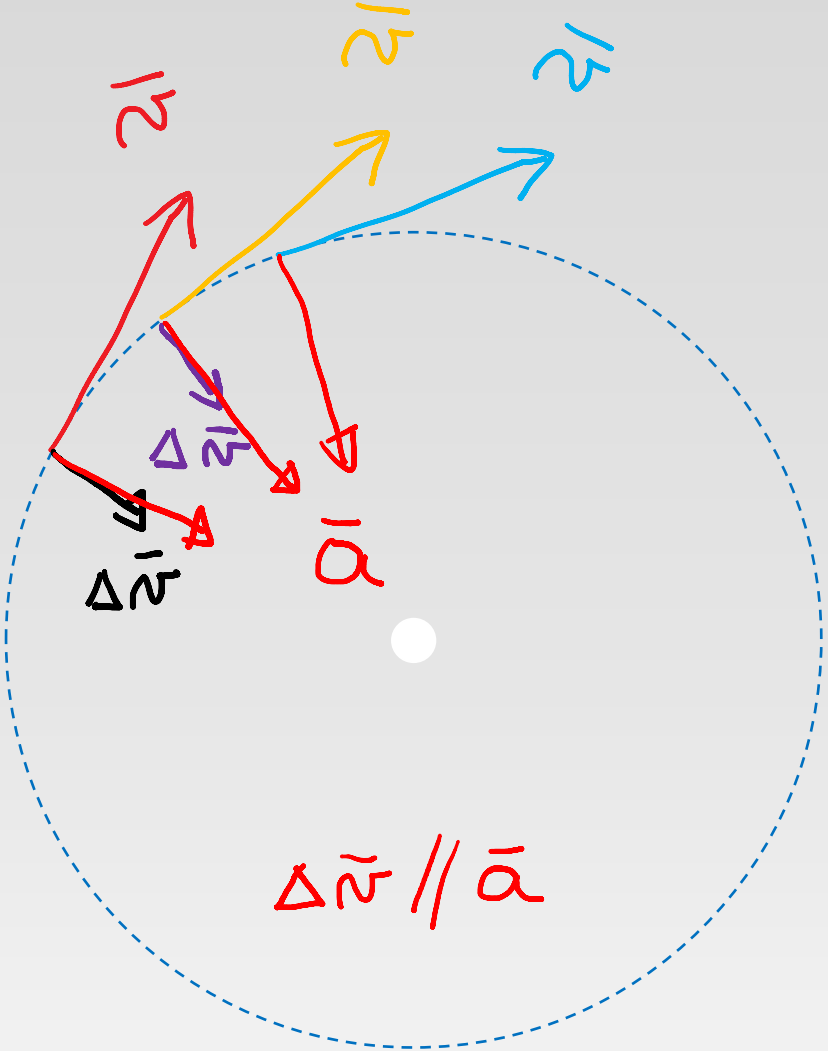
$$|\vec{V}| = \frac{2\,\pi R}{T} = wR$$



Esta relación es válida para cualquier movimiento circular.



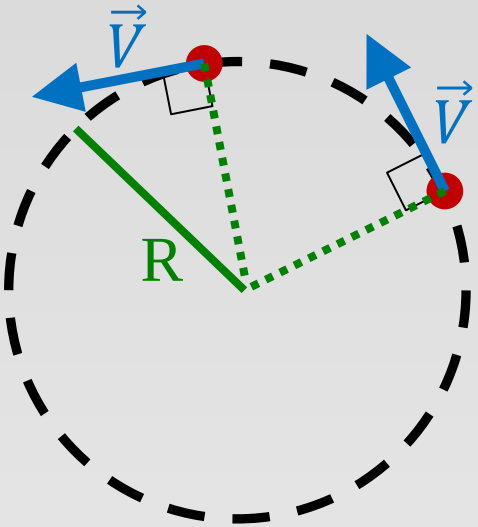




En un movimiento circular uniforme, la aceleración apunta hacia el centro. La llamamos aceleración centrípeta

Movimiento circular uniforme

$|\vec{V}|$= cte, pero $\vec{V} \neq$ cte



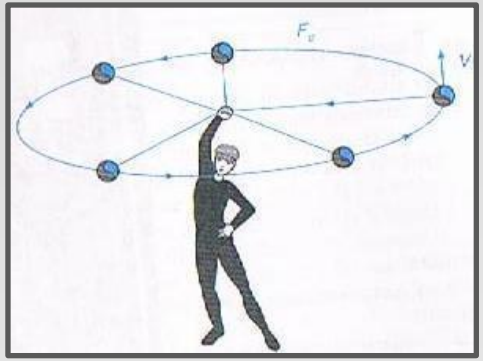
¡Existe una aceleración!

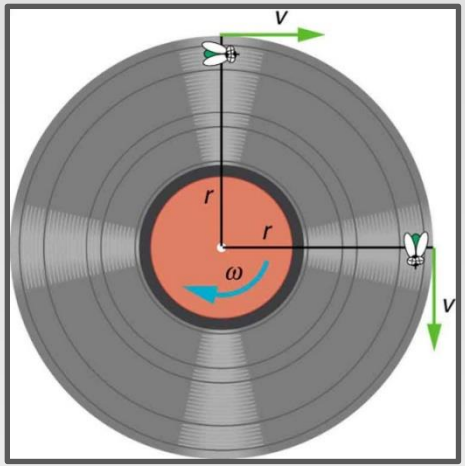
Apunta hacia el centro y se llama aceleración centrípeta:

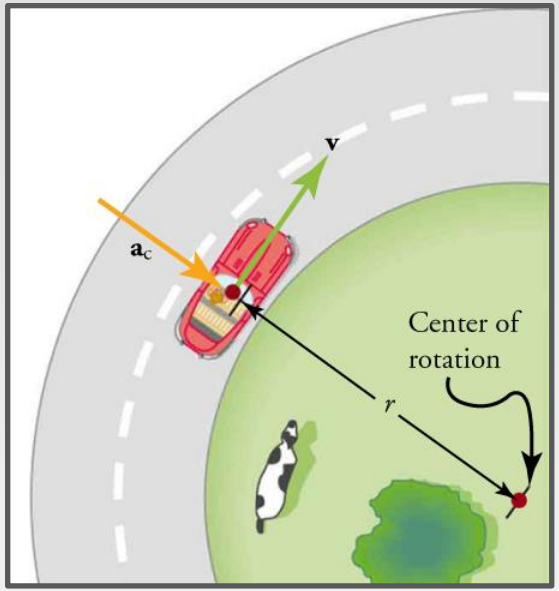
$$|\overrightarrow{a_c}| = \frac{V^2}{R} = w^2 R$$

Aplicamos la segunda ley de Newton en la dirección radial:

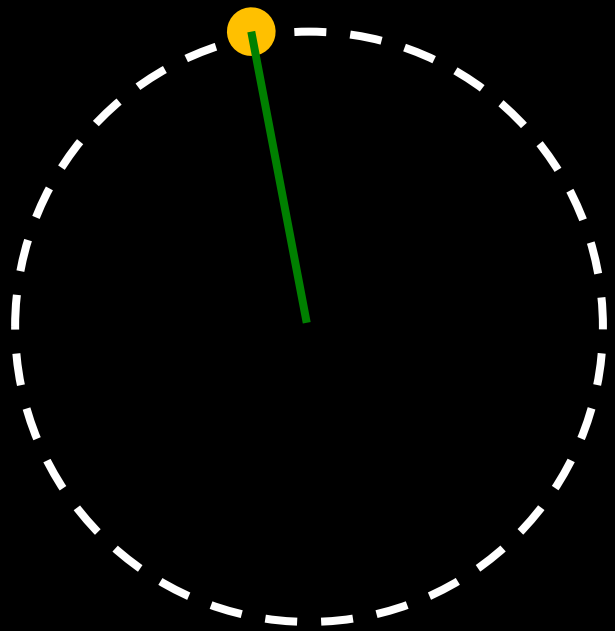
$$\sum F_{Rad} = m\, a_c = m\, \frac{V^2}{R}$$







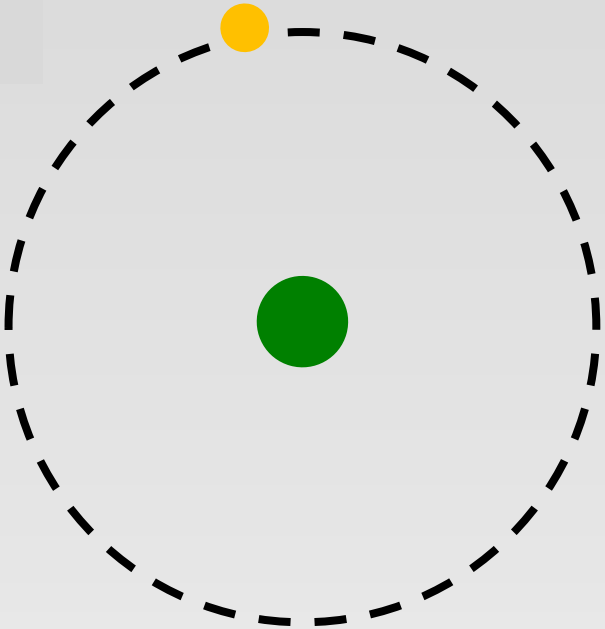
Pelotita atada a un hilo



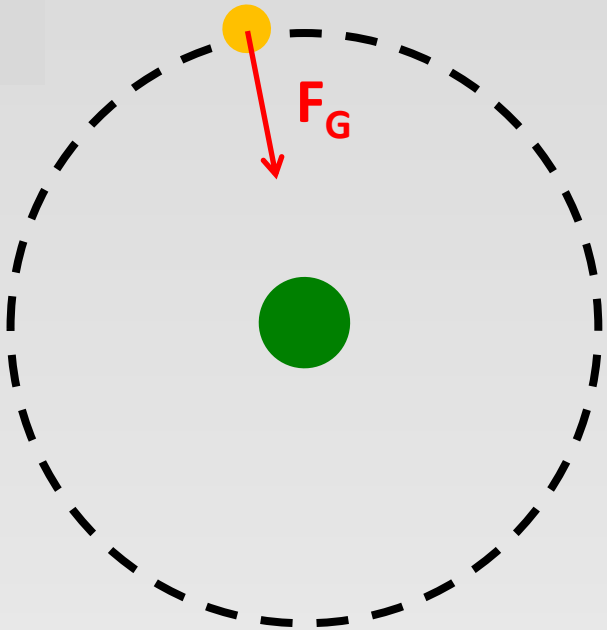
Lanzamiento de martillo:

https://www.youtube.com/watch?time_continue=30&v=VTTl_dW714Q&feature=emb_logo

Orbita de un satélite geosíncrono



$$|F_G| = \frac{G M_T m_s}{R^2}$$



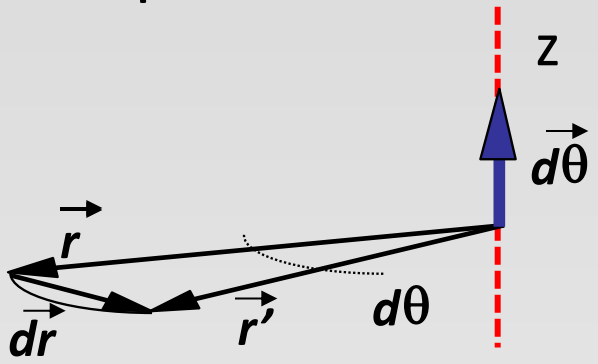
$$\Sigma F_R = \frac{G M_T m_s}{R^2} = m_s a_c = m_s \frac{v^2}{R}$$

$$v = \left(\frac{G M_T}{R}\right)^{1/2} \qquad T = 24\ hs$$

$$v = \frac{2\pi R}{T} = \left(\frac{G M_T}{R}\right)^{1/2} \rightarrow R = \sqrt[3]{\frac{G M_T T_{24hs}}{(2\pi)^2}}$$

$$\approx 42.000\ km/h$$

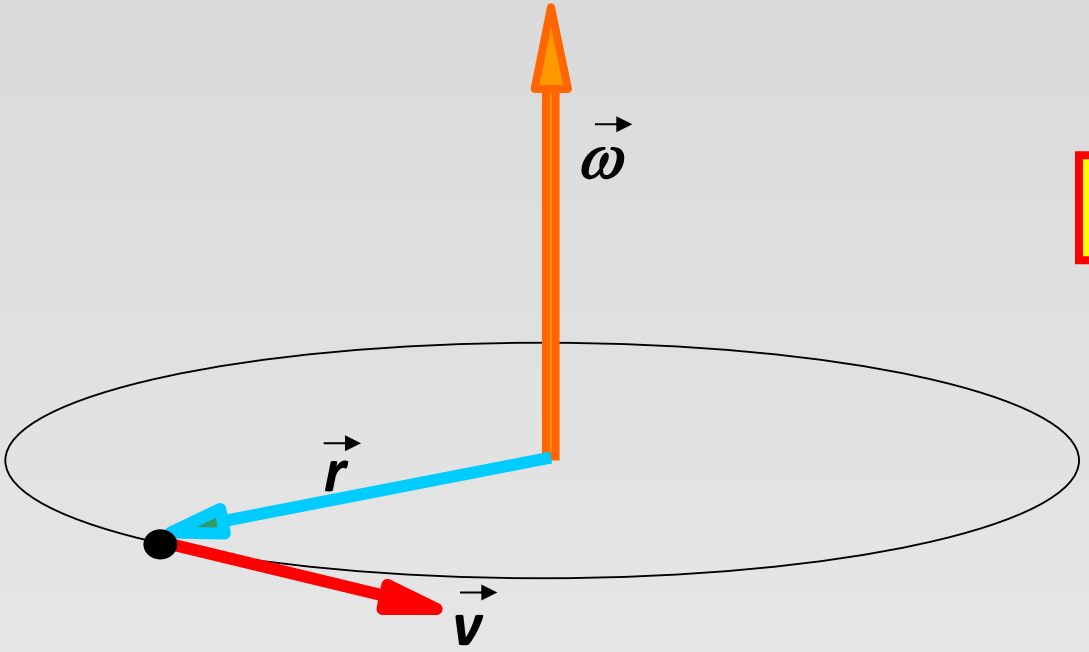
Supongamos que una partícula realiza un movimiento circular en un plano alrededor de un eje fijo “z”



$$d\vec{\mathbf{r}} = d\vec{\theta} \times \vec{\mathbf{r}}$$

$$\vec{v} = \frac{d\vec{\mathbf{r}}}{dt} = \frac{d\vec{\theta} \times \vec{\mathbf{r}}}{dt} = \frac{d\vec{\theta}}{dt} \times \vec{\mathbf{r}} = \vec{\boldsymbol{\omega}} \times \vec{\mathbf{r}}$$

donde definimos la velocidad angular como: $\vec{\omega} = \dfrac{d\vec{\theta}}{dt}$



$$\boxed{\vec{v} = \vec{\omega} \times \vec{\mathbf{r}}}$$

$$v = |\ \vec{\mathbf{v}}\ | = |\ \vec{\omega} \times \vec{\mathbf{r}}\ | = |\vec{\omega}\ | \cdot |\ \vec{\mathbf{r}}\ |\ \ sen 90° = |\vec{\omega}\ | \cdot |\ \vec{\mathbf{r}}\ | = \omega\ r$$

¿Qué pasa cuando $\boldsymbol{\omega}$ varía con el tiempo?

Definimos la **aceleración angular** como:

$$\boxed{\vec{\alpha} = \frac{d\vec{\omega}}{d\mathbf{t}}}$$

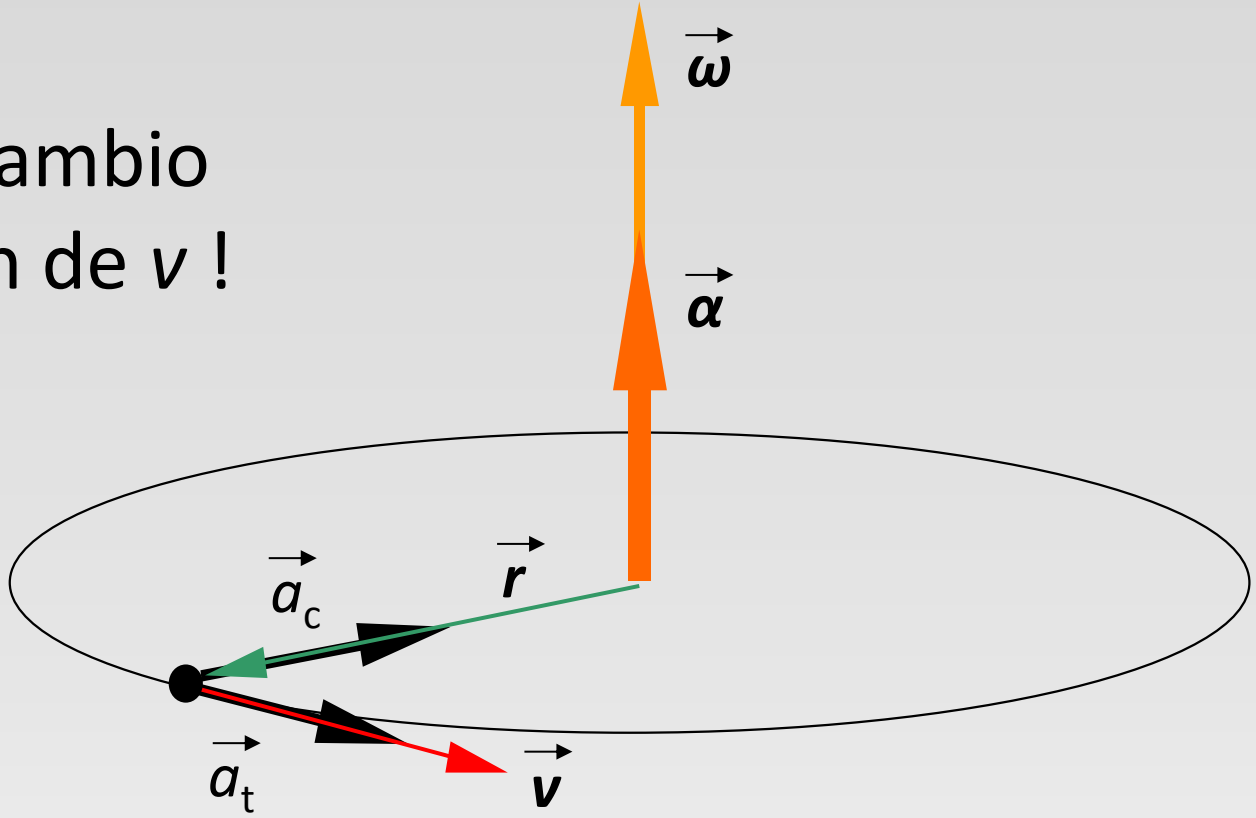
¿Qué pasa con la aceleración $\vec{a}$ cuando $\vec{w}$ varía con el tiempo?

$$\vec{a} = \frac{d\vec{\mathbf{v}}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{\mathbf{r}}) = \frac{d\vec{\omega}}{dt} \times \vec{\mathbf{r}} + \vec{\omega} \times \frac{d\vec{\mathbf{r}}}{dt} = \underbrace{\vec{\alpha} \times \vec{\mathbf{r}}} + \underbrace{\vec{\omega} \times \vec{\mathbf{v}}}$$

aceleración tangencial!!

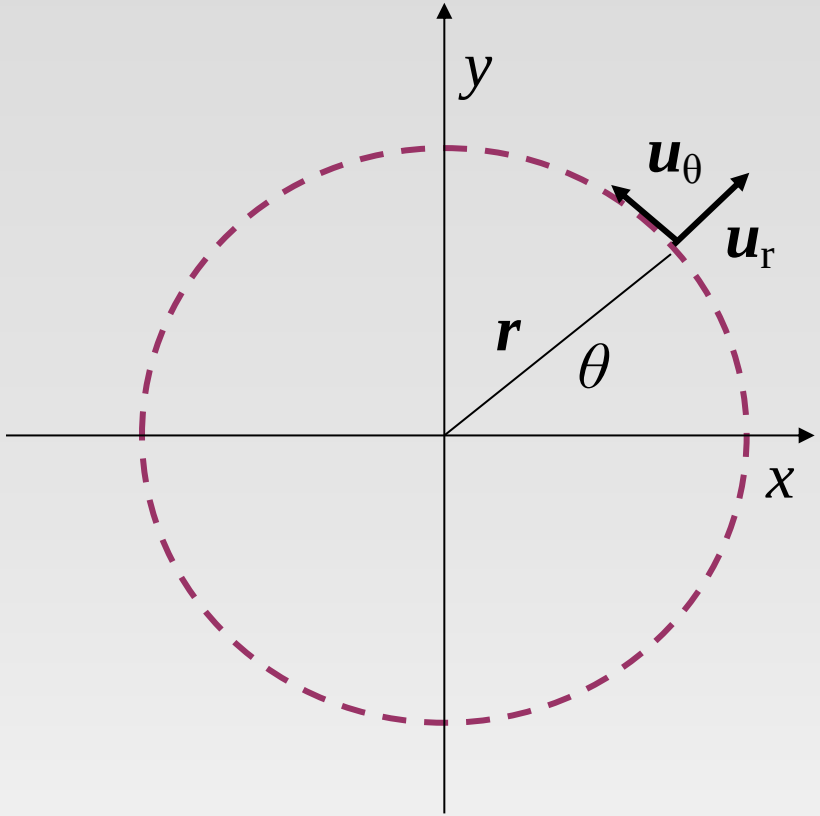
aceleración centrípeta o normal!!

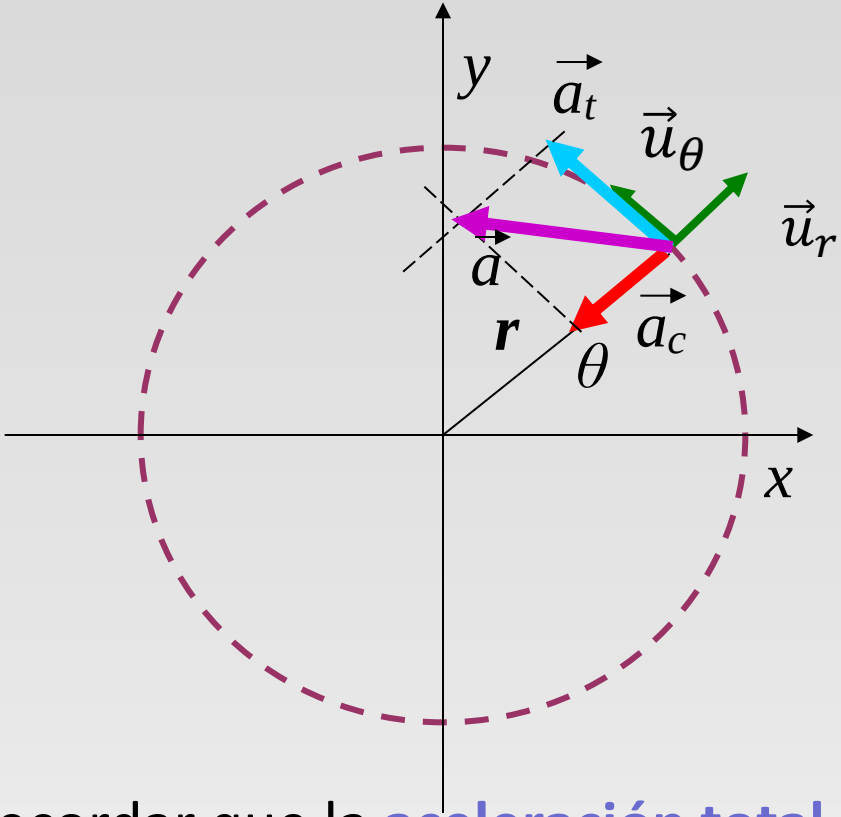
$\vec{a}_c$ origina el cambio de la dirección de v !



$\vec{a}_t$ origina el cambio del módulo de $\boldsymbol{v}$!!

Sistema de coordenadas instantáneo





$$\vec{a}_c = -\frac{v^2}{r}\vec{u}_r$$

$$|\vec{a}_c| = \frac{v^2}{r}$$

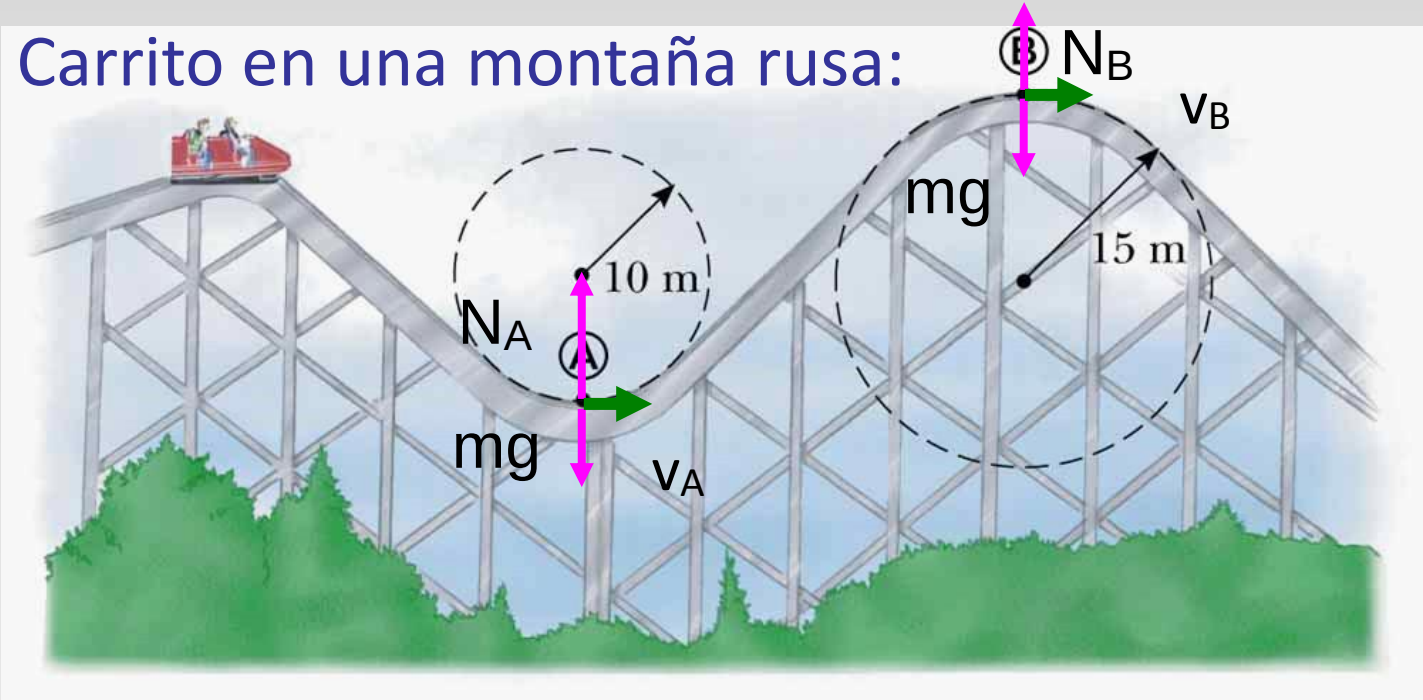
$$\vec{a}_t = r\ \alpha\ \vec{u}_\theta$$

$$|\vec{a}_t| = r\ \alpha$$

$$a = \sqrt{a_t^2 + a_c^2}$$

Recordar que la **aceleración total** ($\vec{a}$) es la suma **vectorial** de las aceleraciones **tangencial** ($\vec{a}_t$) y **centrípeta** ($\vec{a}_c$) !!!

Carrito en una montaña rusa:



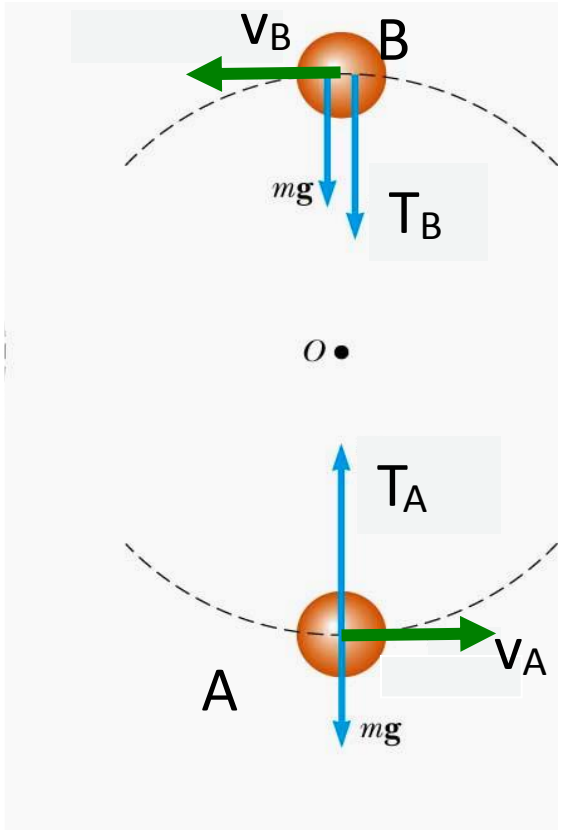
En A: $$\sum F_y = N_A - mg = m\, a_{c,A} = m\,\frac{v_A^2}{R_A}$$

En B: $$\sum F_y = mg - N_B = m\, a_{c,B} = m\,\frac{v_B^2}{R_B}$$

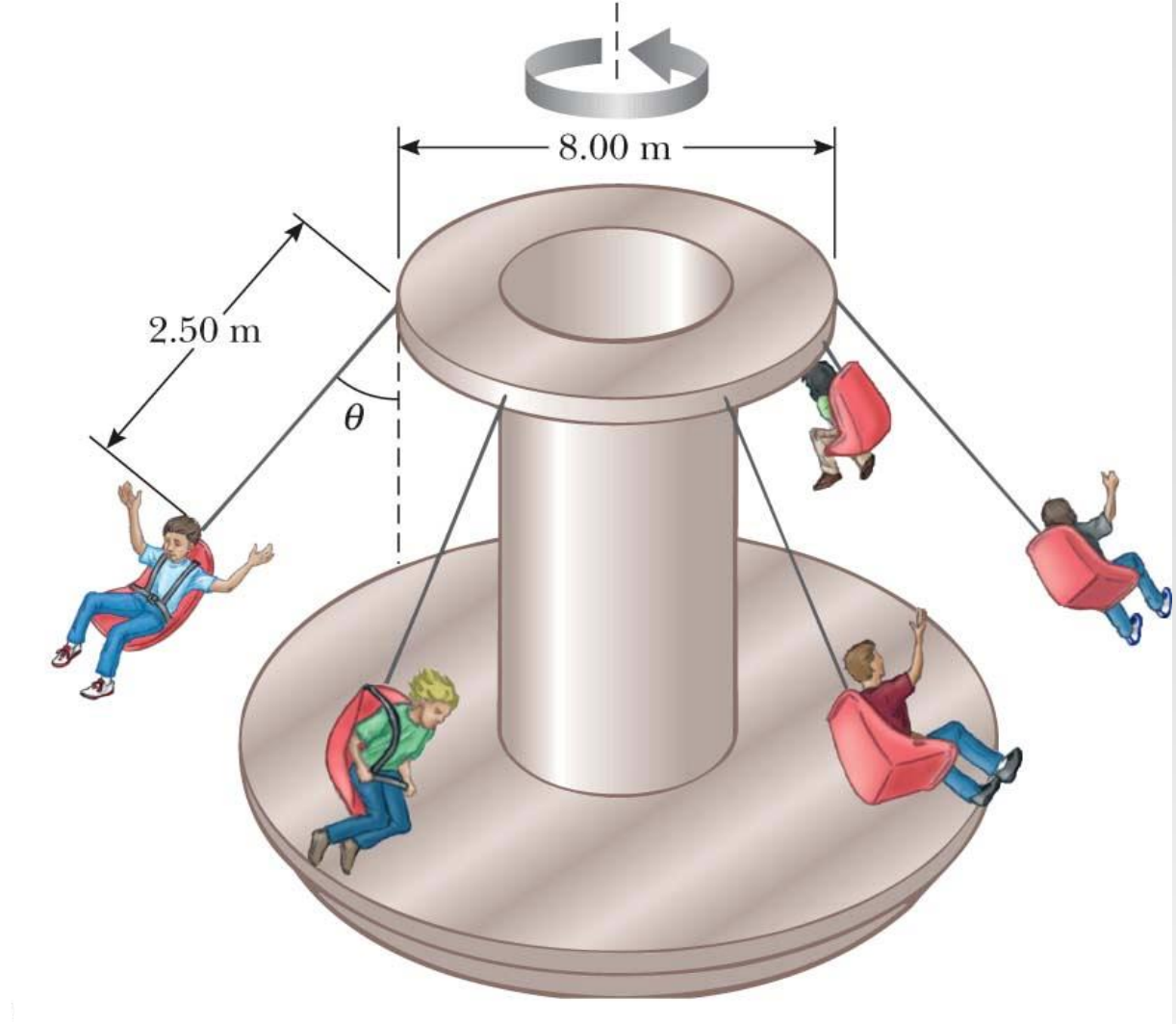
Pelota atada a un hilo, rotando en un círculo vertical:

En B: $$\sum F_y = T_B + mg = m\ a_c = m\frac{v_B^2}{R}$$

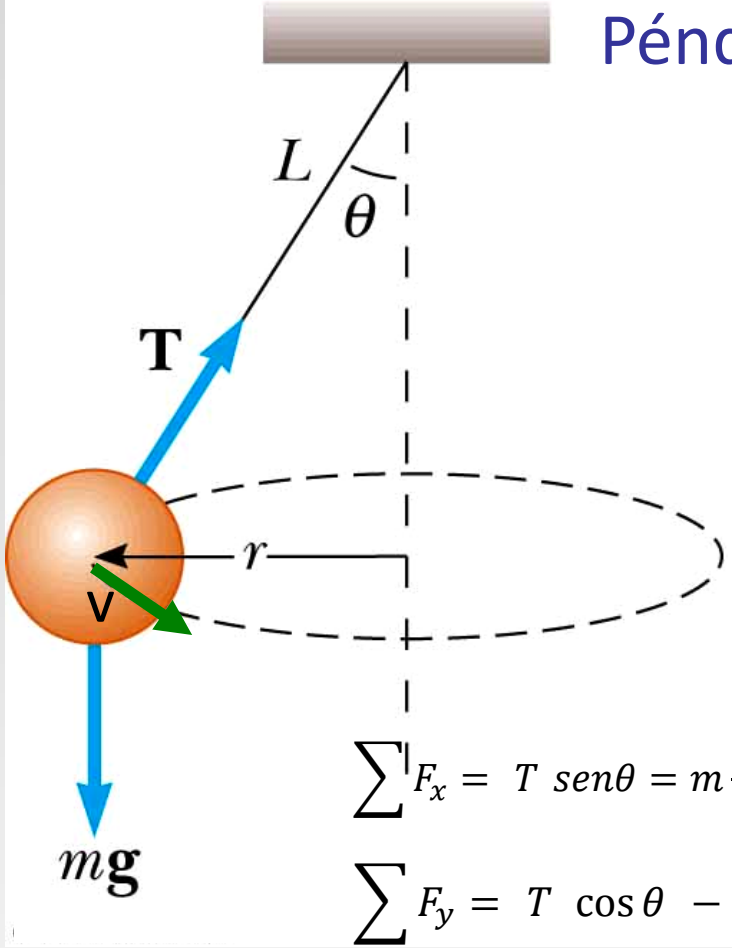
En A: $$\sum F_y = T_A - mg = m\ a_c = m\frac{v_A^2}{R}$$



Calesita

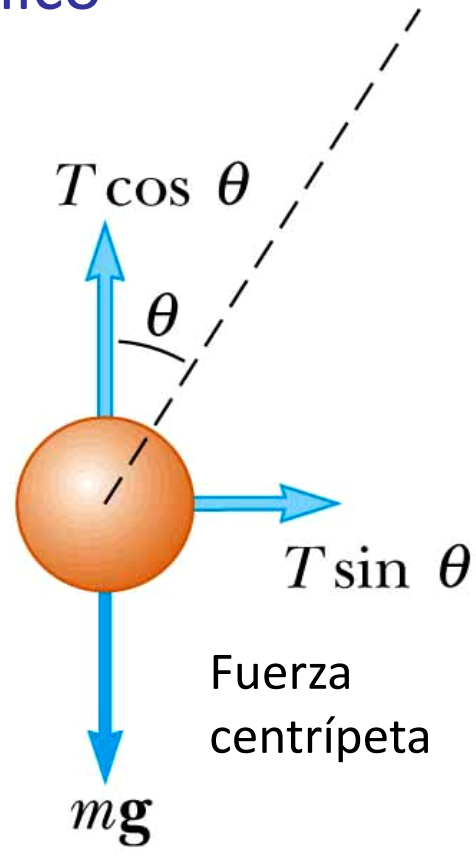


Péndulo cónico

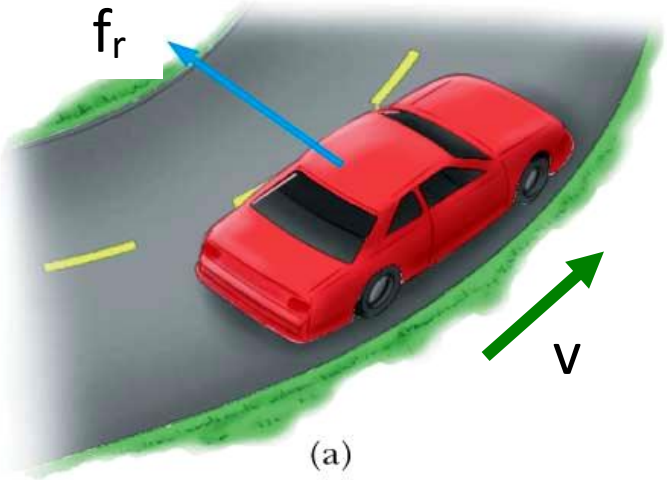


$$\sum F_x = T\ sen\theta = m\frac{v^2}{R}$$

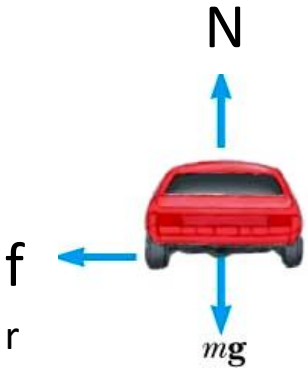
$$\sum F_y = T\ \cos\theta\ - mg = 0$$



Fuerza centrípeta



(a)



(b)

Automóvil tomando una curva sin peralte:

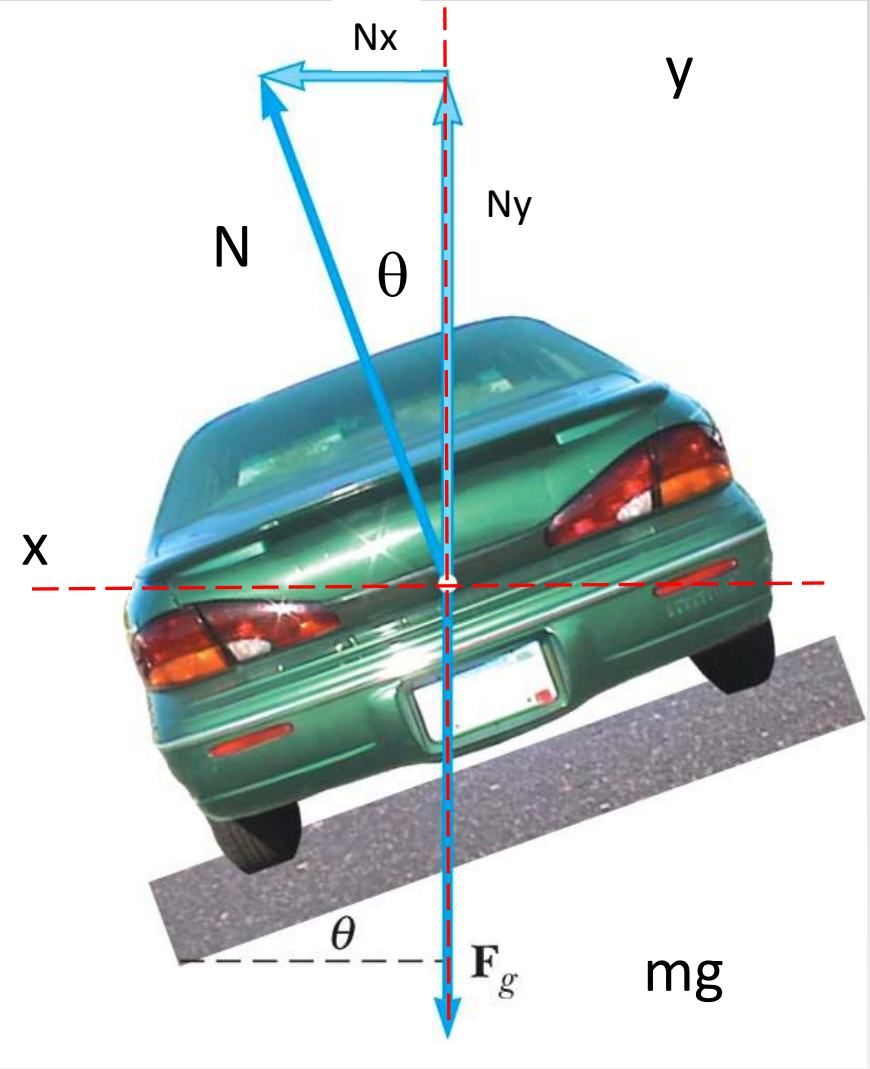
¿qué velocidad máxima puede tomar sin derrapar?

$$\sum F_y = N - mg = 0$$

$$\sum F_x = f_{r,max} = m\,a_c = m\,\frac{v_{max}^{\;2}}{R}$$

$$f_{r,max} = \mu_e\,N = \mu_e\,mg$$

$$\Rightarrow v_{max} = \sqrt{\mu_e\,R\,g}$$



¿A qué velocidad se toma una curva con peralte en ausencia de fuerza de roce?

$$\sum F_y = N_y - mg = 0$$

$$\Rightarrow N\cos\theta = mg$$

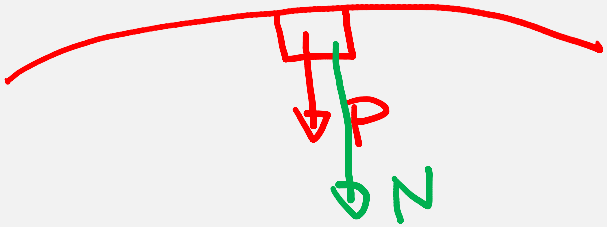
$$\sum F_x = N_x = m\ a_c$$

$$\Rightarrow N\ sen\theta = m\frac{v^2}{R}$$

$$\Rightarrow\ tg\theta = \frac{v^2}{g\ R} \Rightarrow v_{max} = \sqrt{g\ R\ tg\theta}$$

La rueda de la muerte

https://www.youtube.com/watch?v=V52H1Xo7Joc&feature=youtu.be



$$\Sigma F_R = P + N = m\,a_c = m\frac{v^2}{R}$$

$v \uparrow \qquad N \uparrow \qquad ¿\, v_{min}\,?$

$$mg + N = m\frac{v^2}{R}$$

$$v^2 = gR + \frac{N}{m}$$

$$v_{min} = \sqrt{gR} \approx 2{,}6\ m/s$$